

Asset Pricing Fundamentals II: Capital Allocation

MAF307

Equities and Investment Analysis

Lecture 7



Roadmap

Investment Environment (Topic 1 and Topic 2)

Equity Analysis and
Valuation

Portfolio Theory and Applications

Macro &
Industry
Analysis
(Topic 3)

Financial
Statement
Analysis
(Topic 4)

Equity
Evaluation
Models
(Topic 5)

Understanding
Risk
(Topic 6)

Capital
Allocation
(Topic 7)

Optimal Risky
Portfolio
(Topic 8)

CAPM, APT and
factor models
(Topic 9)

Portfolio
Performance
Evaluation
(Topic 10)



Learning Objectives

You'll learn about:

- ❑ The framework of creating Investment Portfolio
- ❑ The Concept of Capital Allocation & Complete Portfolio
- ❑ Risk Free Asset and its proxies
- ❑ Capital Allocation Line (CAL) & its application in identifying optimal complete portfolio an investor



Portfolio Decisions

Capital allocation decision:

Choice of investing in money market or capital market instruments

Asset allocation decision:

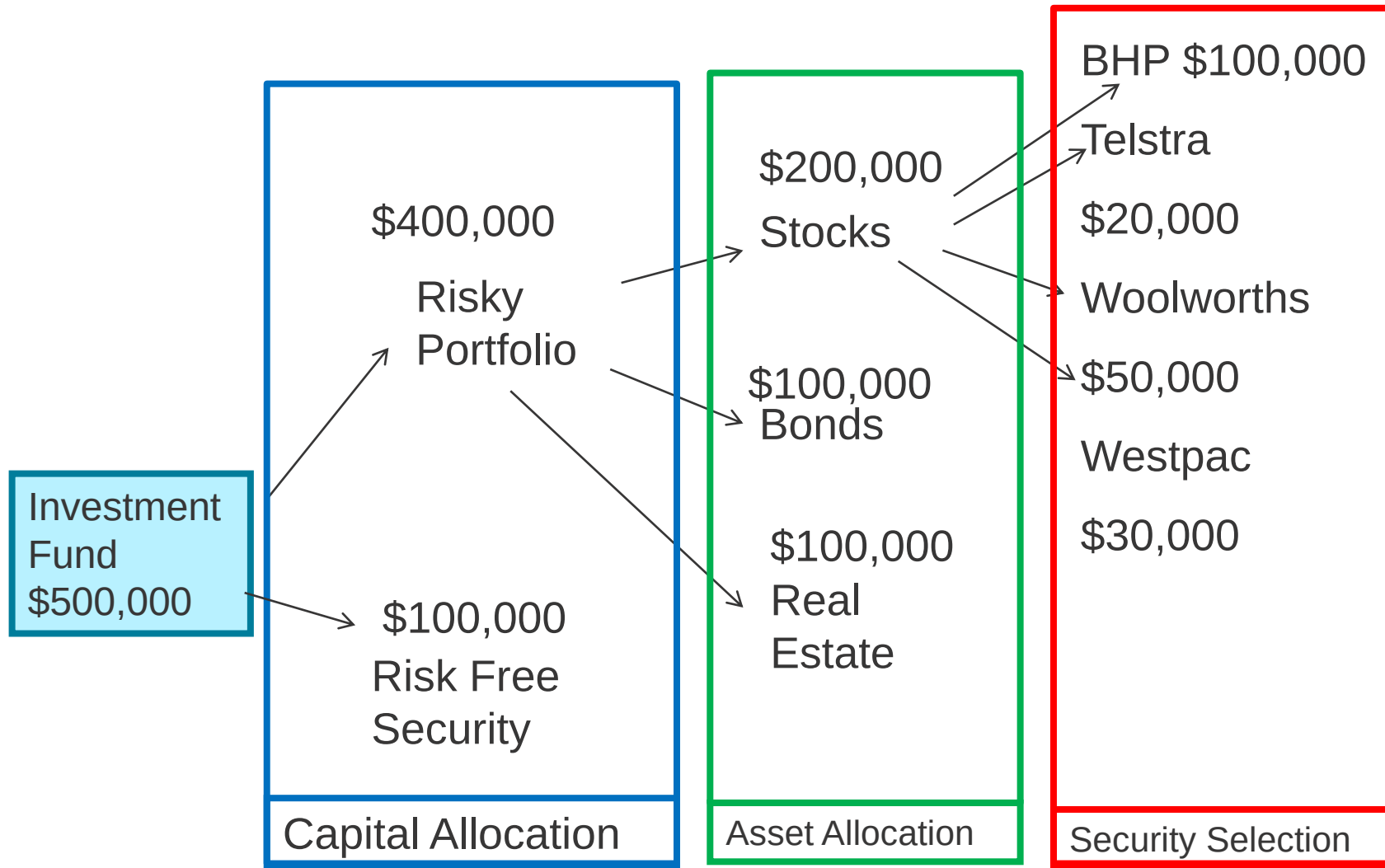
Choosing among various risky assets

Security selection decision:

Choice of particular securities within each asset category



Portfolio Decisions



Capital Allocation

Managing risk return trade-off by splitting investment funds between safe and risky assets

Risk free asset: proxy - Treasury bills , Treasury notes, Treasury bonds

Risky asset: stock (or a portfolio of stocks)

Manage investment risk & return through distribution of wealth between risk free asset and risky portfolio without changing composition of the risky portfolio



Capital Allocation - Example

Allocation of \$1M

More Risky

\$ 200,000 – Risk free security

\$ 800,000 – Risky portfolio

40% Bond

60% Stock

Less Risky

\$ 450,000 – Risk free security

\$ 550,000 – Risky portfolio

40% Bond

60% Stock



What is the “Risk Free Rate” ?

Treasury Bills or Treasury Bonds are treated as the risk free assets – why ?

Government Securities

- Government can issue default free bonds
- What about inflation ?

Is default free perfectly inflation indexed bond – risk free ?

- No still there is interest rate risk

How about T - Bills ?

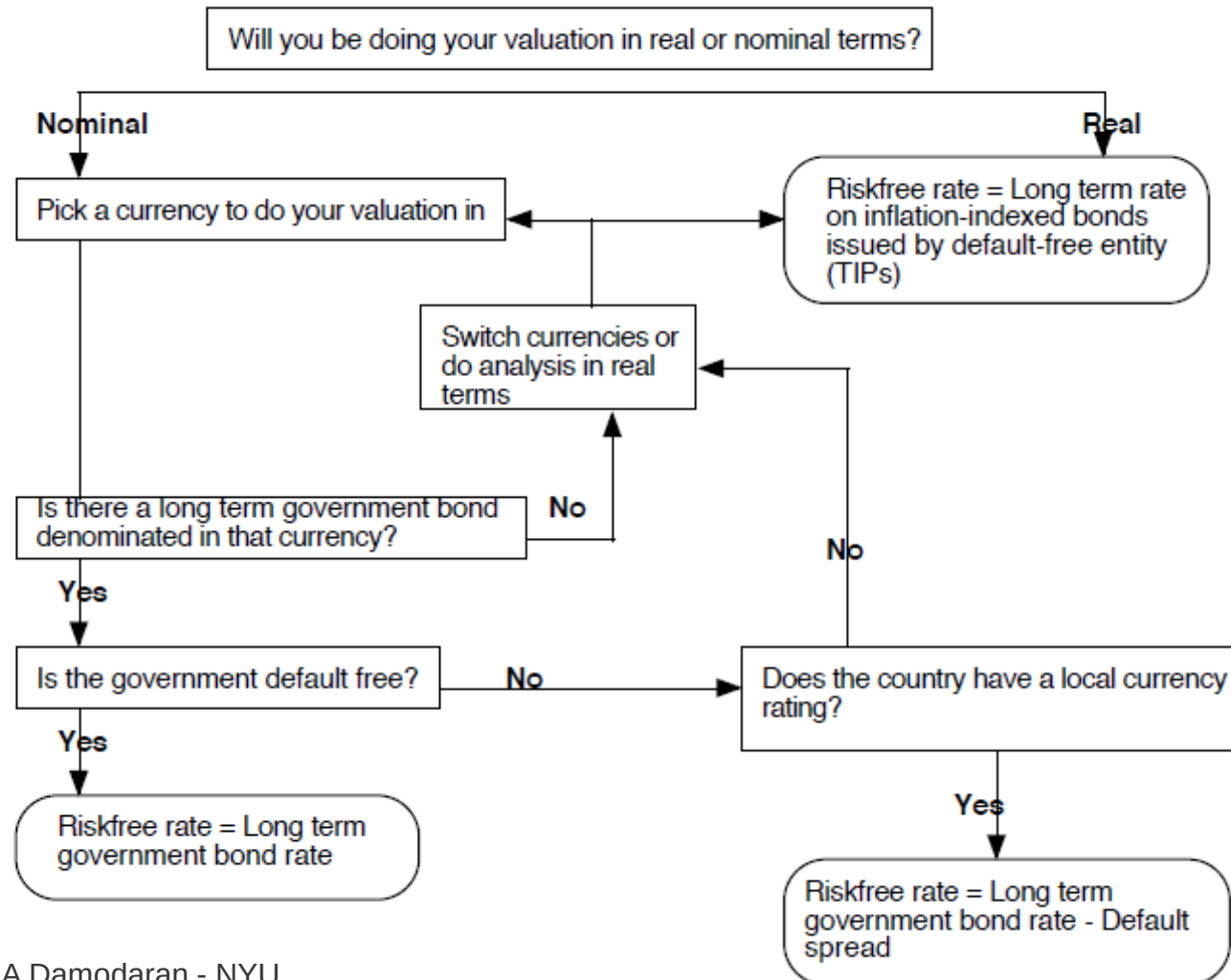
- Short term nature
- Negligible inflation uncertainty
- Insensitive to interest rate fluctuations

But T-Bills have reinvestment risk ?



Risk Free Rate for Long Term Investors

Figure 7: A Framework for estimating Riskfree Rates

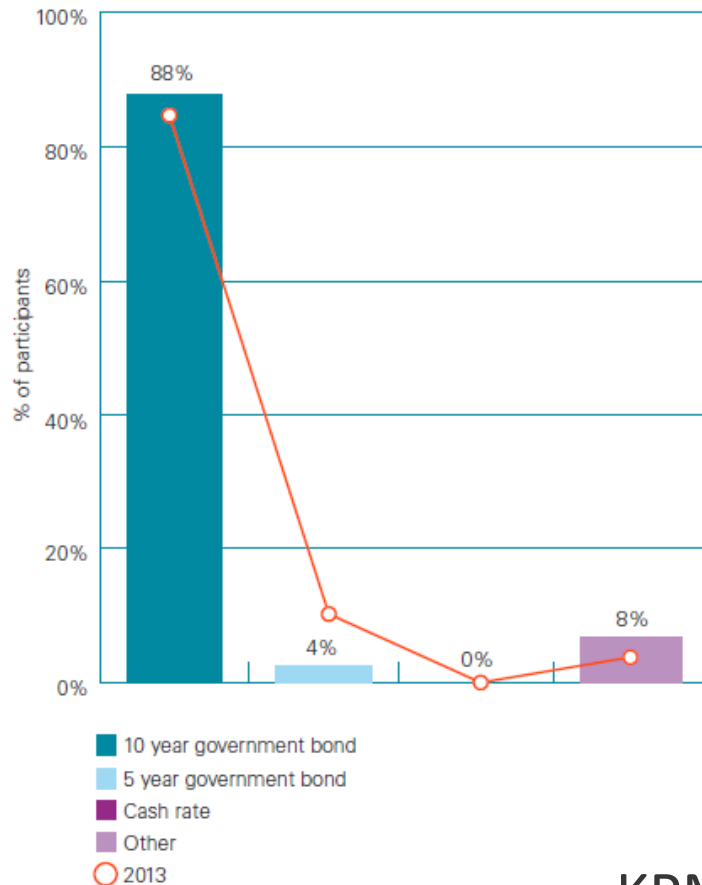


Source: Prof. A Damodaran - NYU

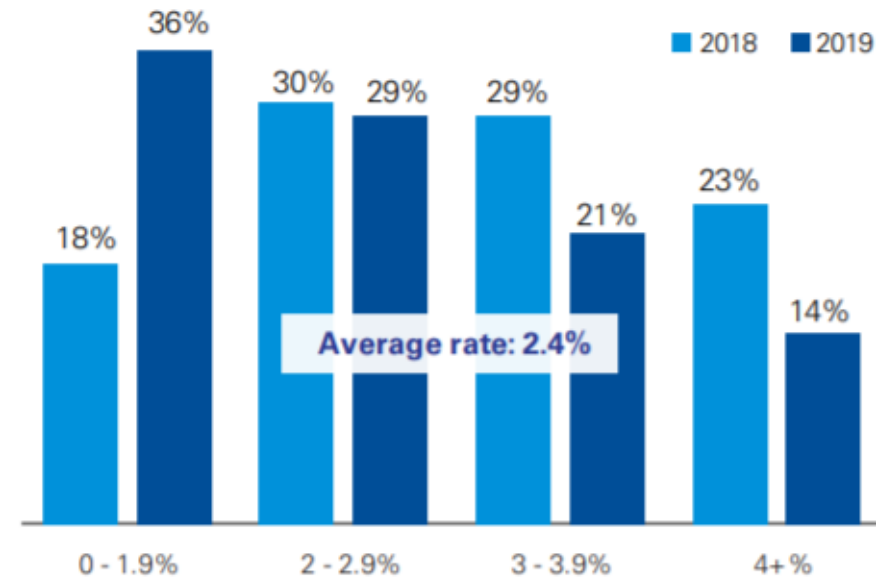


Risk Free Rate – In Practice

Figure 19: **Survey 2015** – Which of the following are generally used as a benchmark for the risk free rate in Australia?



What was the risk free rate you adopted for valuations you completed for a perpetual life Australian asset at 30 June 2019?

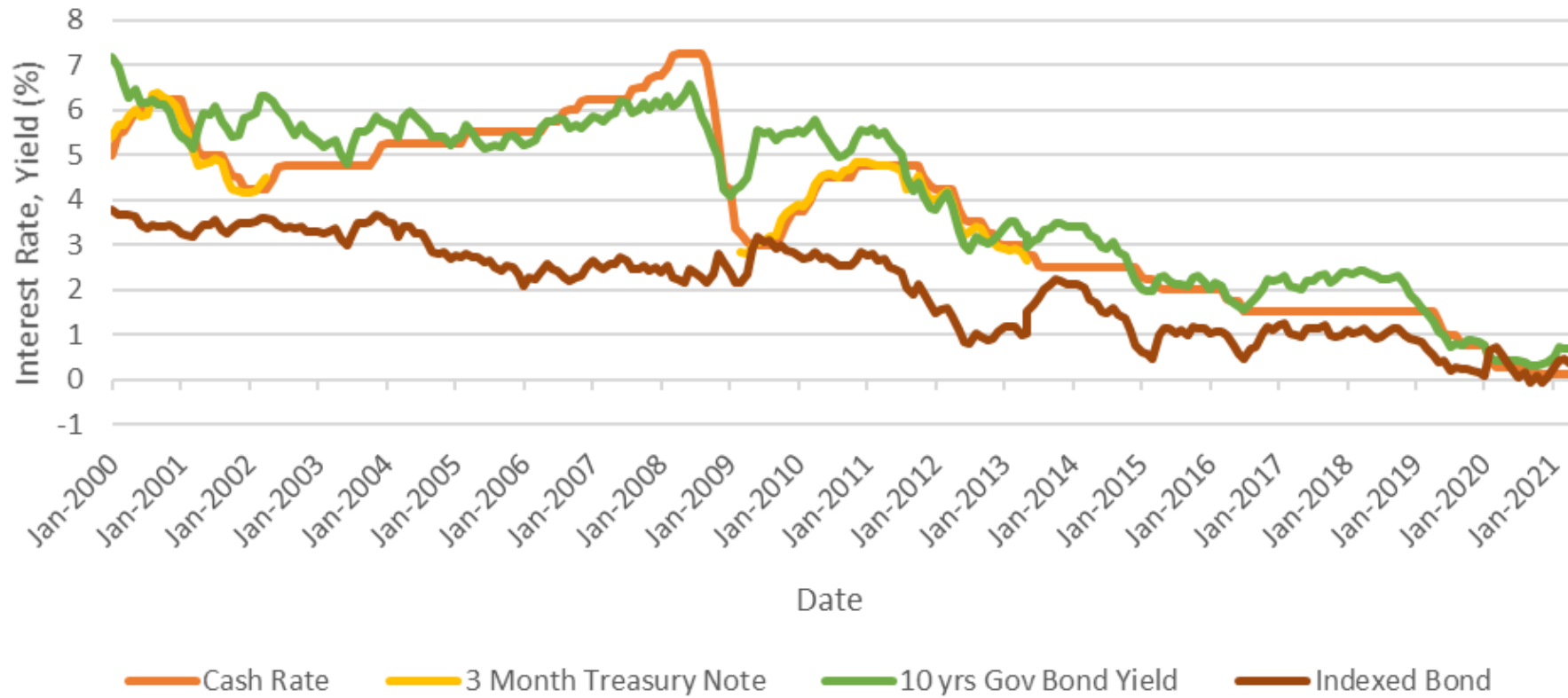


KPMG Survey 2015, 2019

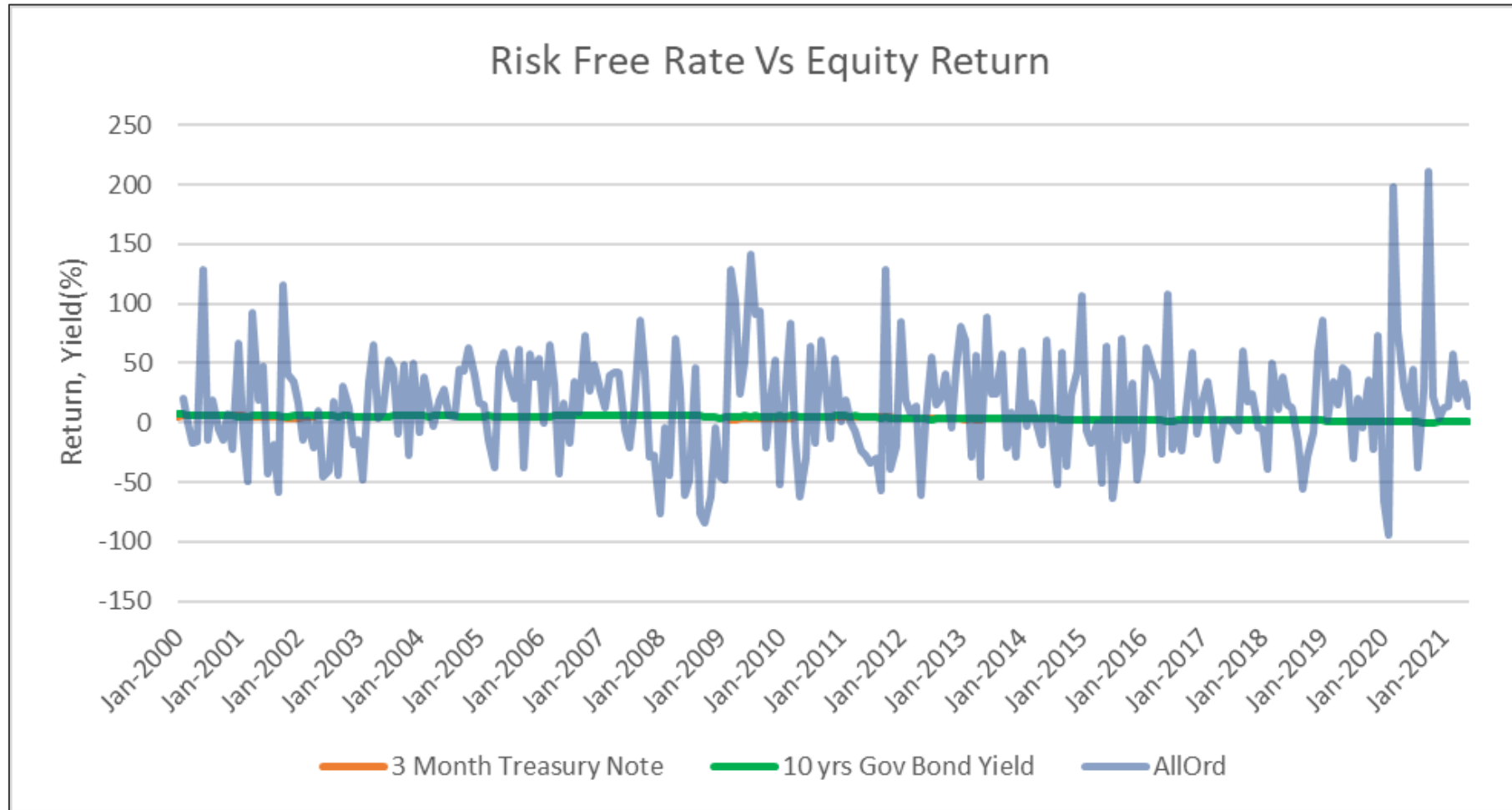


Risk Free Rates

Risk Free Rate - Australia



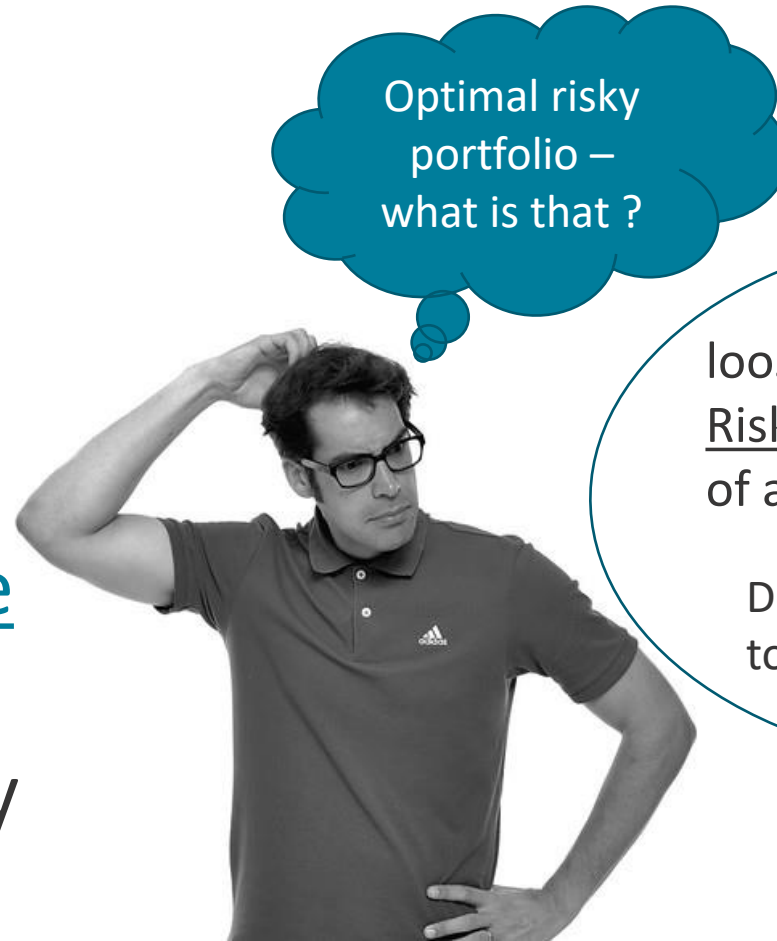
Risk Free Rate Proxies Vs Equity Return



Basic Objective – Capital Allocation

Allocate our funds so that the portfolio suits our risk return preference.

We want to create a Complete Portfolio i.e. portfolio of a risk free asset and an optimal risky portfolio



Optimal risky portfolio – what is that ?

loosely speaking, Optimal Risky Portfolio is the best mix of all risky asset in the market

Don't worry we'll learn how to find it in the next lecture



Portfolio Return, Risk and Investment Preference

W_1 = Proportion of funds in Security 1 , r_1 = Expected return on Security 1, σ_1^2 = Variance of Security 1

W_2 = Proportion of funds in Security 2, r_2 = Expected return on Security 2, σ_2^2 = Variance of Security 2

$\text{Cov}(r_1 r_2)$ = **Covariance** of returns for Security 1 and Security 2

ρ_{12} = **Correlation** of returns for Security 1 and Security 2

Return

$$r_p = W_1 r_1 + W_2 r_2$$

Risk

$$\begin{aligned}\sigma_p^2 &= w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2W_1 W_2 \text{Cov}(r_1 r_2) \\ &= w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2W_1 W_2 \sigma_1 \sigma_2 \rho_{12}\end{aligned}$$

Utility function

$$U = E(r) - 0.005 A \sigma^2$$



Expected Complete Portfolio Return

If we invest “y” percent of our wealth in optimal risky portfolio and rest i.e. (100-y) percent wealth in risk free asset then

The expected complete portfolio return is :

$$E(r_c) = yE(r_p) + (1-y)r_f$$

$$E(r_c) = yE(r_p) + r_f - yr_f$$

$$E(r_c) = r_f + y[E(r_p) - r_f]$$



Don't worry about the calculations. If you are not mathematically inclined, just focus on the formula in **RED box**.

Expected complete portfolio return = Risk free rate + y (Risk premium)



Complete Portfolio - Example

You expect 12% return from your investment

Optimal risky portfolio gives 15% return with 22% standard deviation.

Risk free rate is 7%

How should you allocate your capital to earn expected return of 12%?

We know, $E(r_c) = r_f + y(r_p - r_f)$ so,

$$12\% = 7\% + y(15\% - 7\%) \quad \text{Or} \quad y = 5/8 = 0.625 = 62.5\%$$

We need to invest 62.5% in the risky portfolio



Complete Portfolio Risk

$$\sigma_c^2 = w_1^2 \sigma_p^2 + w_2^2 \sigma_{rf}^2 + 2w_1w_2 \text{Cov}(r_p, r_f)$$

So,

$$\sigma_c^2 = w_1^2 \sigma_p^2$$

$$\sigma_c = w_1 \sigma_p \text{ Or } \sigma_c = \gamma \sigma_p$$

From previous example:

$$\sigma_c = 22 \gamma = 22 * 0.625 = 13.75\%$$



Don't worry about the calculations. If you are not mathematically inclined, just focus on the formula in **RED box**.

Risk Return Trade Off

Expected Return $E(r_c) = r_f + y[E(r_p) - r_f]$

Portfolio Risk

$$\sigma_c = y \sigma_p$$

Or $y = \sigma_c / \sigma_p$

Replacing the value of y in expected return equation –

$$E(r_c) = r_f + \frac{\sigma_c}{\sigma_p} [E(r_p) - r_f]$$



Capital Allocation Line (CAL)

$$E(r_c) = r_f + \frac{[E(r_p) - r_f]}{\sigma_p} \sigma_c$$

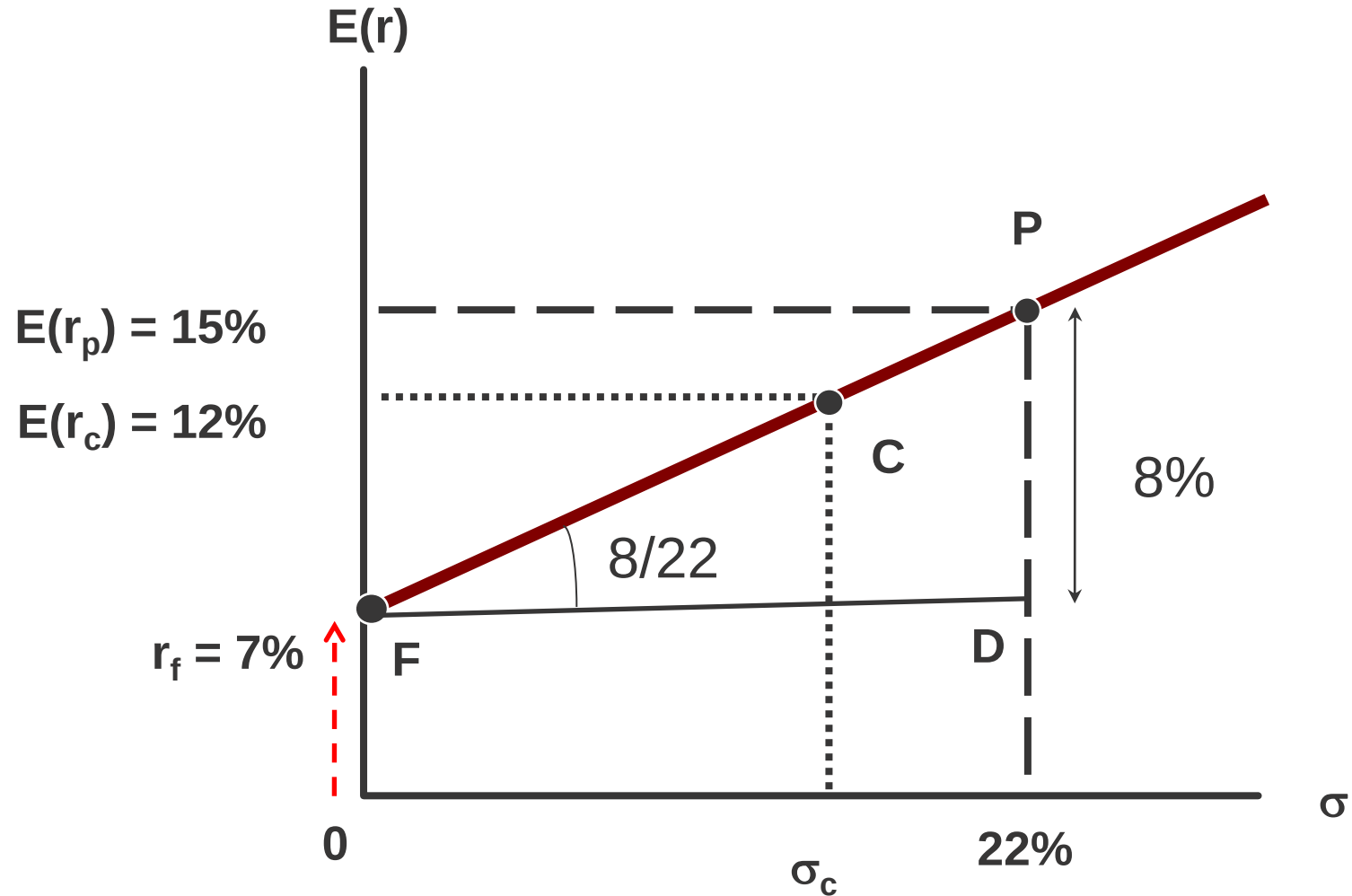
This is the relationship between expected return $[E(r_c)]$ and risk $[\sigma_c]$ of the complete portfolio

This is a linear relationship – which means if we plot $E(r_c)$ on y-axis and σ_c in x axis of a graph, we'll get a straight line

This line is called the Capital Allocation Line (CAL)



Capital Allocation Line



Capital Allocation Line (CAL)

CAL represents the **investment opportunity set**

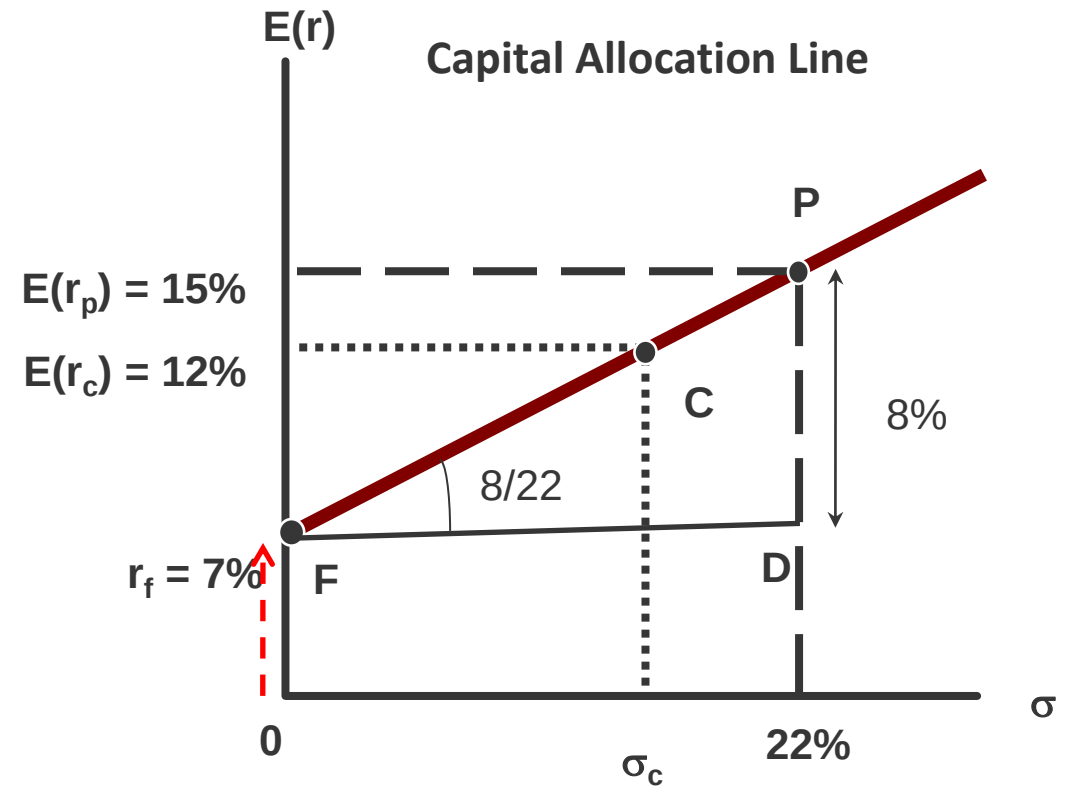
The slope of CAL represents **excess return per unit of risk**

$$S = \frac{[E(r_p) - r_f]}{\sigma_p} = \text{Reward to variability ratio}$$



Capital Allocation

Preference	Capital Allocation	Portfolio Outcome
Do not want any Risk	100% in risk free asset (point F)	7% return, 0% risk
Wants 15% return	100% in optimal risky portfolio (P)	15% return, 22% risk
Wants 12% return	62.5% in optimal risky portfolio (C)	12% return, 13.75% risk



How can we earn more than 15% ?

Leverage

How can we earn more than 15% ? - It is possible with leverage

Leverage:

- Increasing the investable funds by borrowing
- borrowing to invest in risky portfolio

Example - margin trading

Pros - High expected return

Cons - High standard deviation (risk)



Leverage Example

You have \$100,000 for investment

And you borrow another \$50,000 at risk free rate

Invest total \$150,000 in the risky portfolio

You have borrowed 50% of your own wealth

Borrowing is = Negative investment

So in your portfolio you have -50% investment on risk free asset and 150% investment in risky asset



Portfolio with Leverage

Borrow at the Risk-Free Rate and invest in risky portfolio

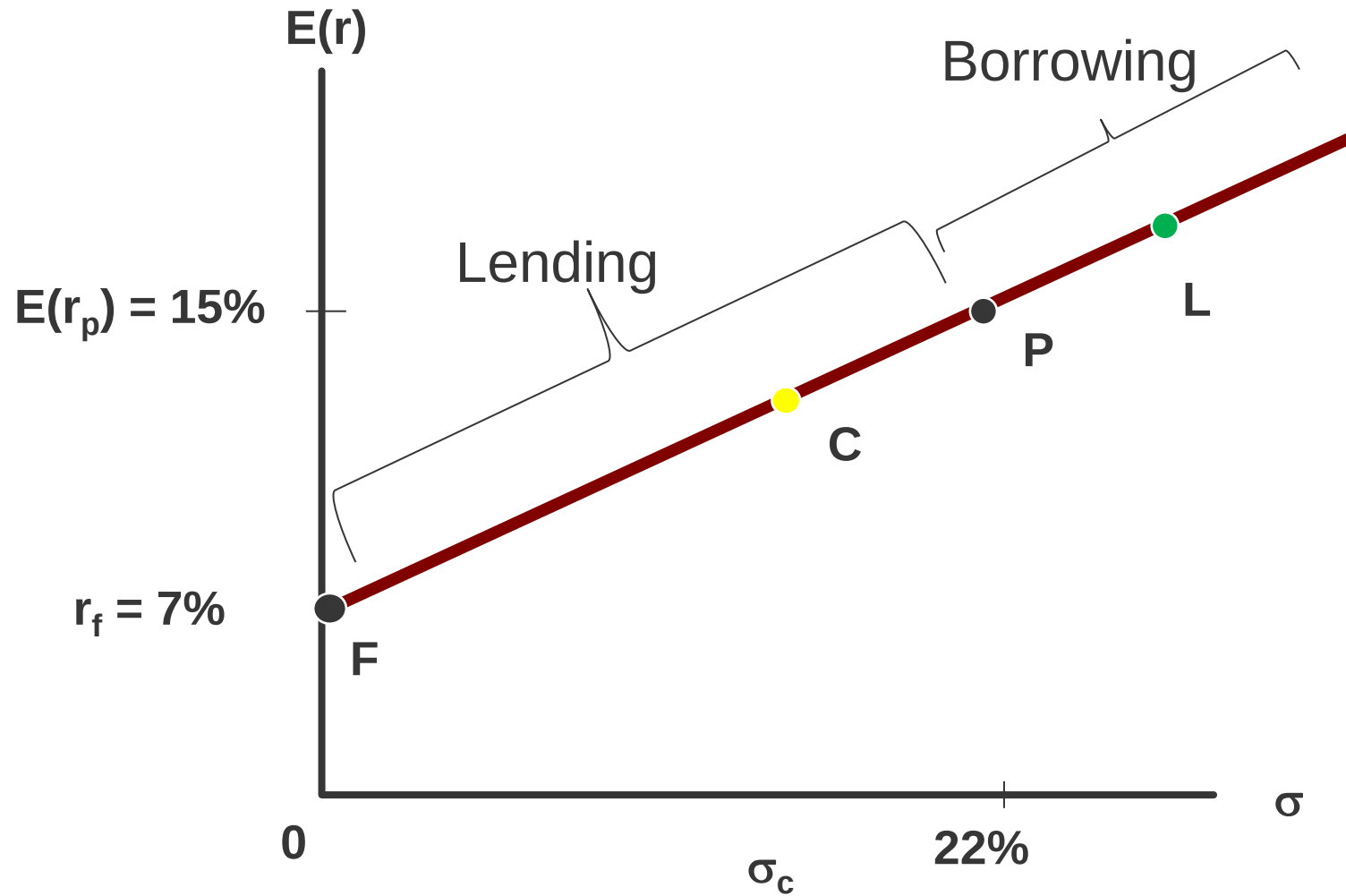
Using the example of 50% Leverage -

$$E(r_c) = r_f + y[E(r_p) - r_f] = (.07) + (1.5)(0.15 - 0.07) = 0.19$$

$$\sigma_c = y \sigma_p = (1.5)(0.22) = 0.33$$



CAL with Leverage



Leverage

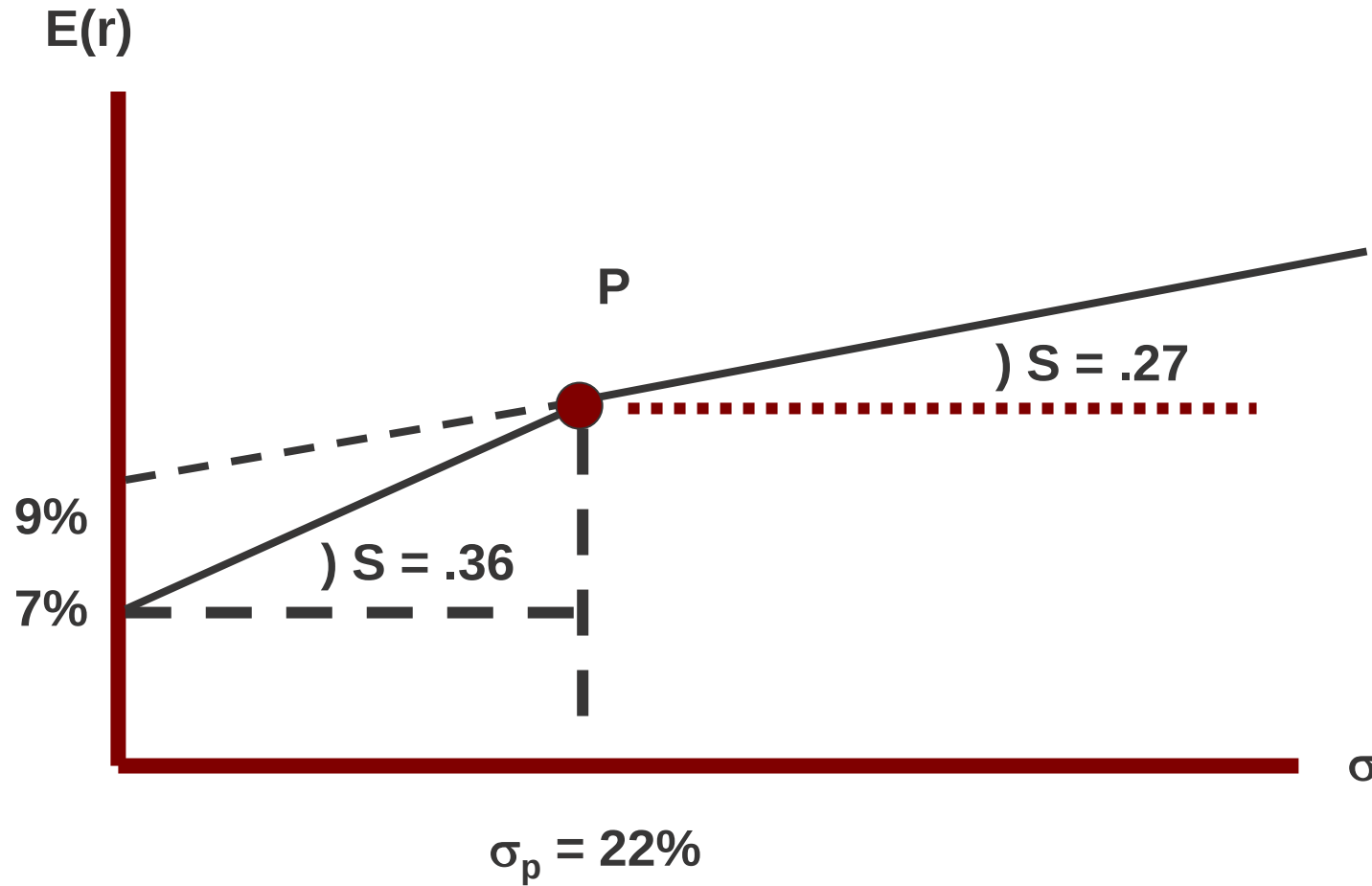
Borrowing for investment at what rate?

At risk-free rate ?

- Everybody can not borrow at risk-free rate
- Higher interest rate for certain borrowers
- Leverage portfolio has an higher standard deviation than unleveraged one



CAL with Higher Borrowing Rate



Risk Aversion and Capital Allocation

Greater levels of risk aversion lead to larger proportions of investment in risk free asset

Lower levels of risk aversion lead to larger proportions of investment in the portfolio of risky assets

Leveraged combinations can provide high levels of returns for those who can tolerate high levels of risk



So how much risk should I take ?

Depends on your risk - return preferences

Your utility function can help you answer this question

$$U = E (r) - .005 A \sigma^2$$

Where:

U = utility

$E (r)$ = expected return on the asset or portfolio

A = coefficient of risk aversion

σ^2 = variance of returns



Optimal Capital Allocation

In a Complete Portfolio

$$E(r_c) = r_f + y[E(r_p) - r_f]$$

$$\sigma_c^2 = y^2 \sigma_p^2$$



Don't worry about the calculations. If you are not mathematically inclined, just focus on the formula in **RED box**.

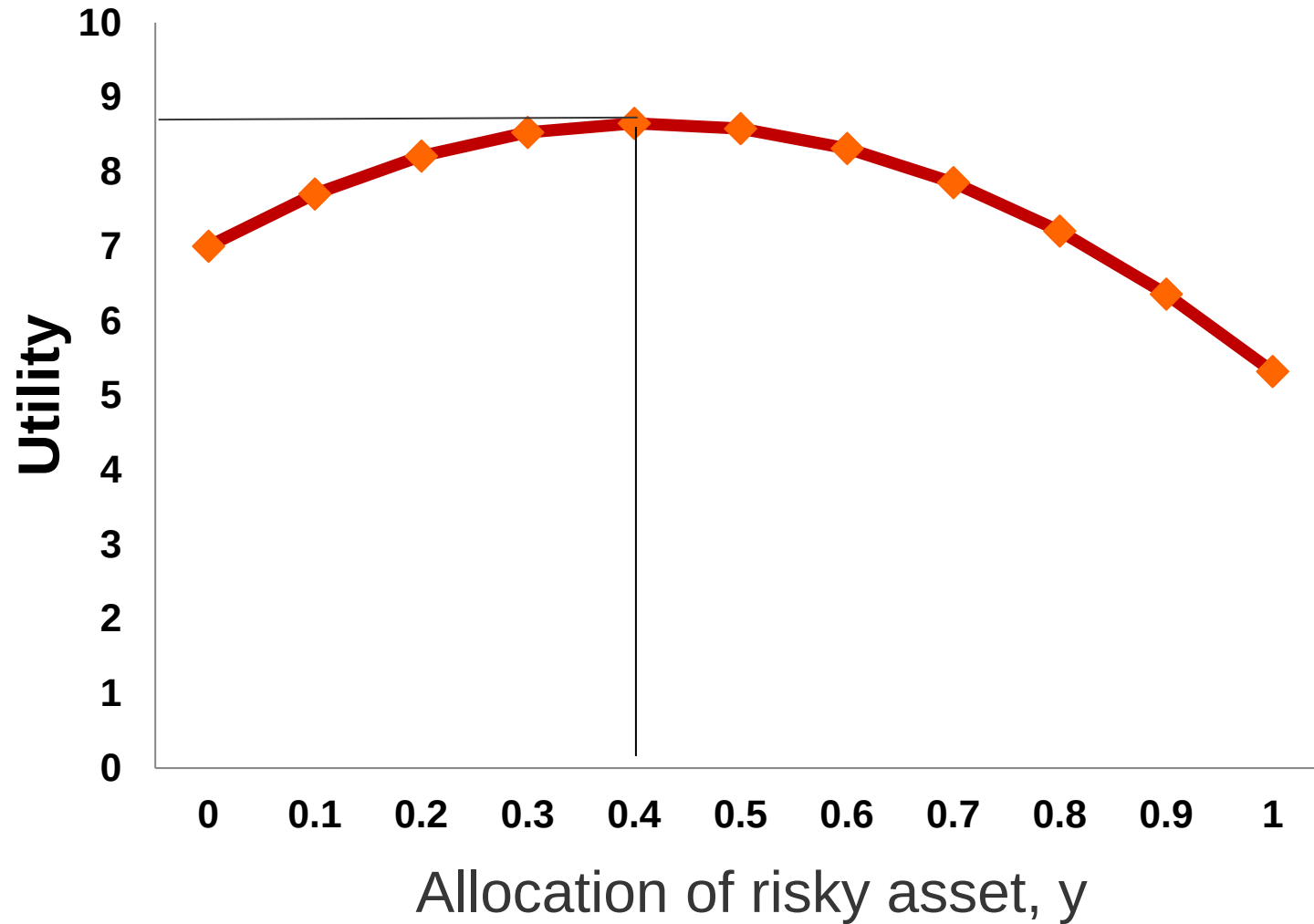
Therefore, utility from investing in the complete portfolio is :

$$U = E(r_c) - 0.005A\sigma_c^2 = r_f + y[E(r_p) - r_f] - 0.005Ay^2\sigma_p^2$$

The value of y that maximizes utility is :

$$y^* = \frac{E(r_p) - r_f}{0.01A\sigma_p^2}$$

Capital Allocation & Investor's Utility



Optimal Capital Allocation

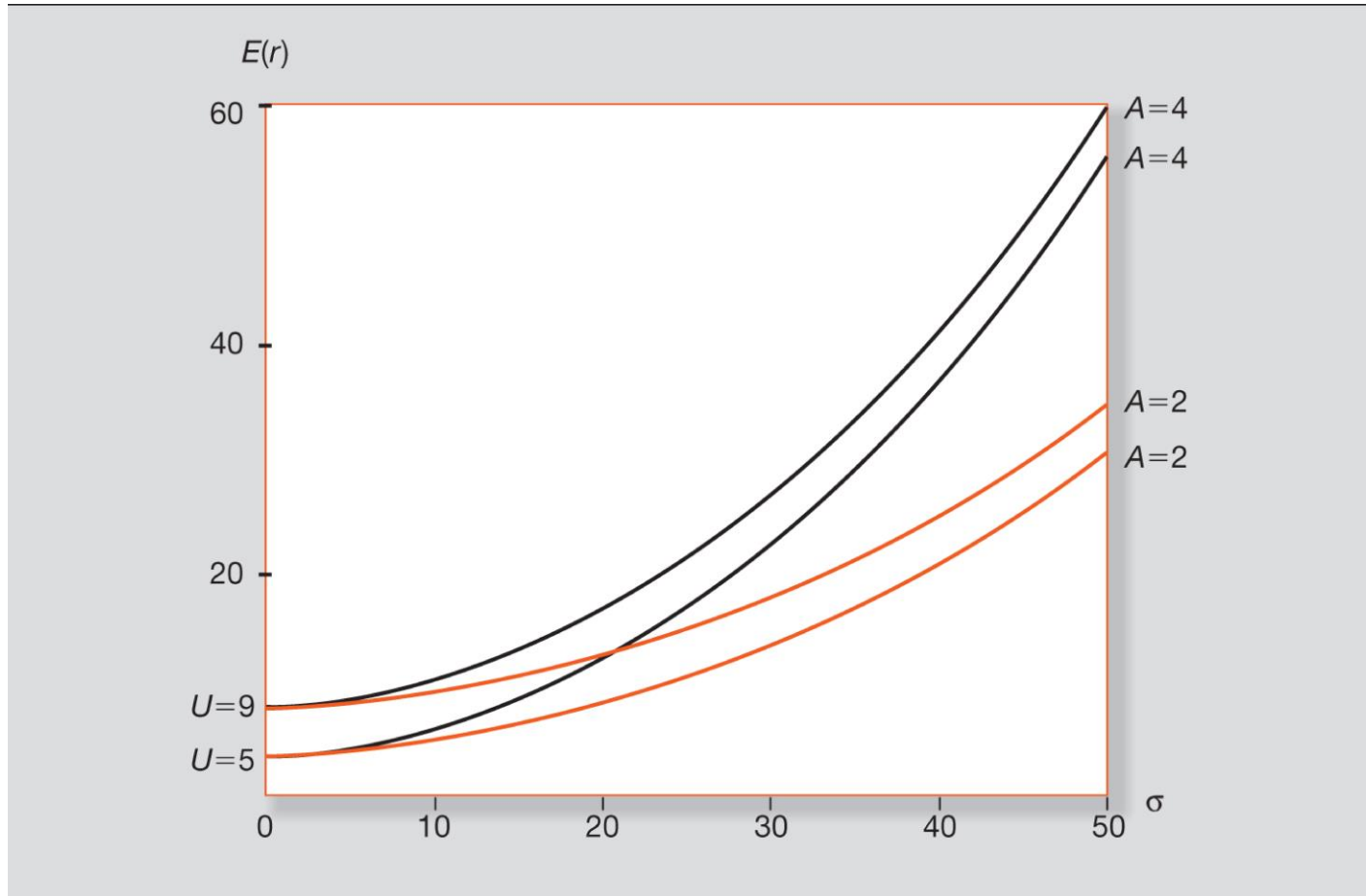
Another way of finding Optimal Capital Allocation

Plot investor's indifference curves along with CAL

The point at which CAL becomes tangent to investor's indifference curve that point is the ideal portfolio



Remember - Indifference Curves ?



- Indifference curves connect investments with equal utility.
- Investors are indifferent to investments on a indifference curve.

Indifference Curves - Recap

Higher Indifference curves higher levels of utility

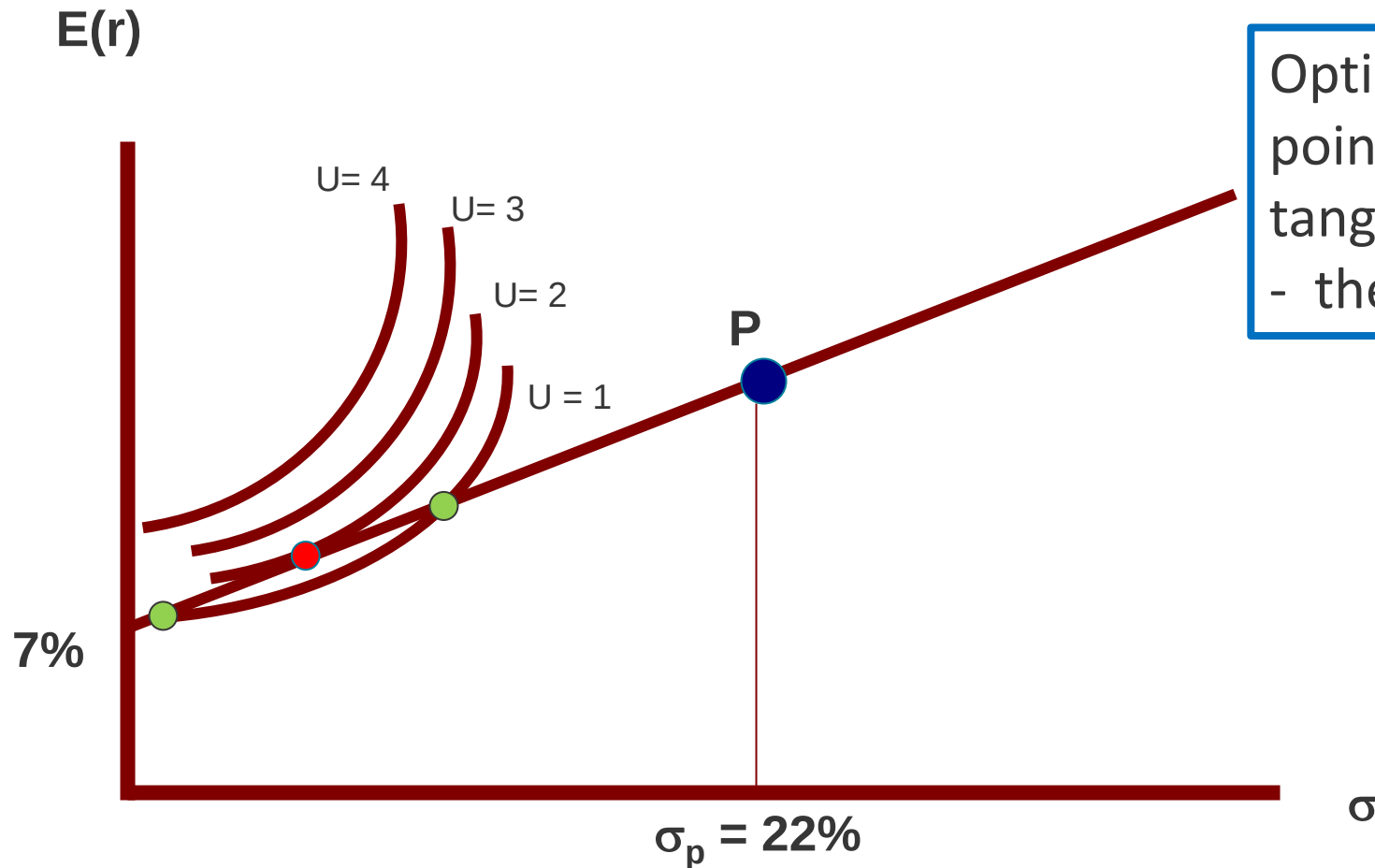
- Assets on higher indifference curve provides - higher expected return for any level of the risk

More risk-averse investors has steeper indifference curves than the less risk-averse investors

- More risk averse investors demand - higher expected return for same amount of risk

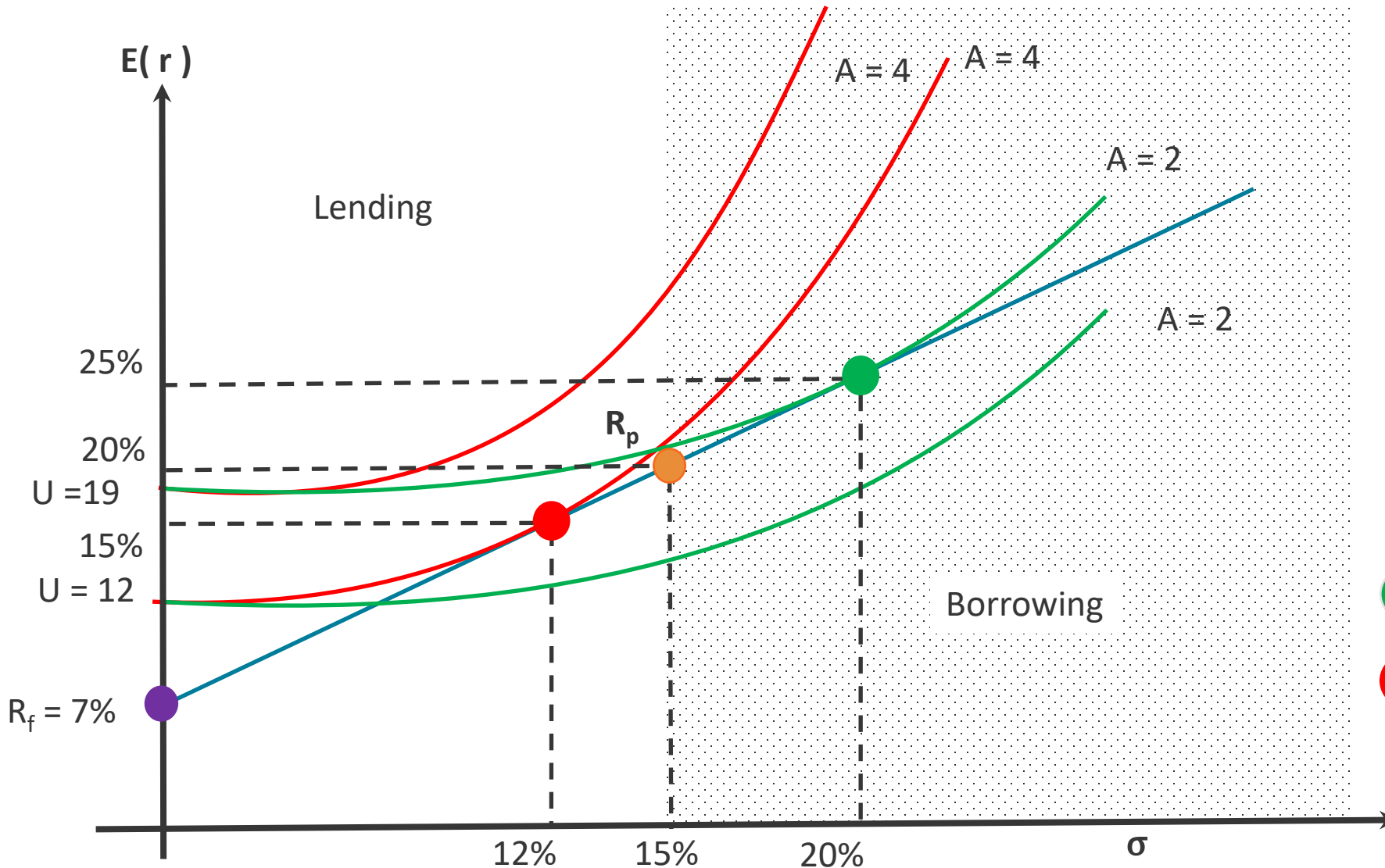


Finding Optimal Capital Allocation



Optimal Complete Portfolio – point where CAL becomes tangent to an indifference curve - the **RED** point on this graph

Finding Optimal Complete Portfolio



Click the buttons to add / delete graph elements

Risk Free Rate



Optimal Risky Portfolio



CAL



Lending / Borrowing



Investor 1 Indifference curve



Investor 2 Indifference Curve



Utility Values



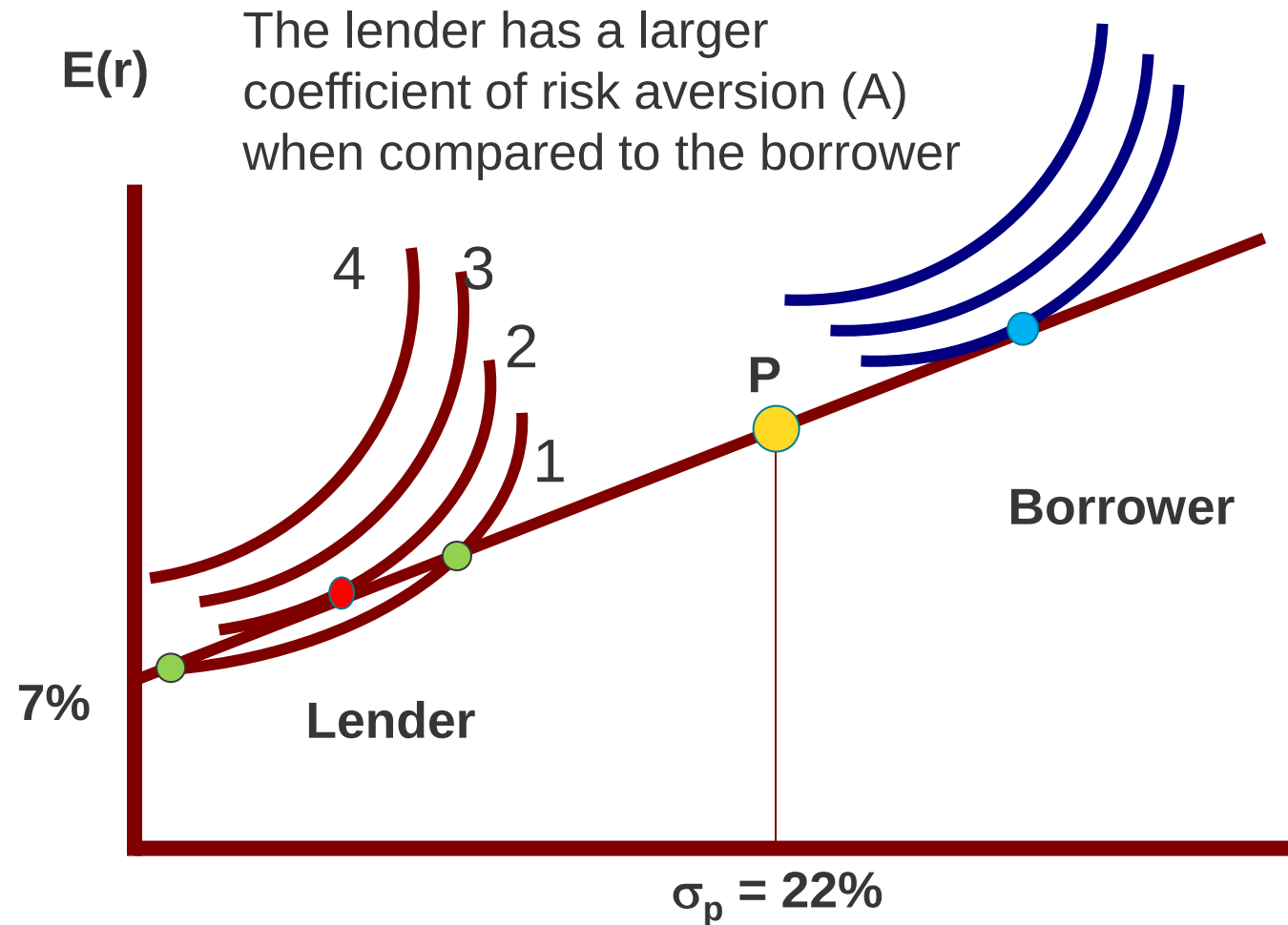
Investor 1 Optimal Complete Portfolio



Investor 2 Optimal Complete Portfolio



CAL with Risk Preferences



Capital Allocation Decision

Investors' decision of capital allocation depends on their risk preferences

Point at which investor's indifference curve becomes tangent to the Capital Allocation Line that point represents the ideal portfolio for that investor



Capital Market Line

CAL is derived with the risk free asset and the optimal risky portfolio

What is this “Optimal Risky Portfolio” ? **(Next week’s discussion)**

How do we determine this ?

Active vs Passive strategy

A natural choice well diversified passive portfolio of stocks – Market Index

Capital Market Line – A special case of CAL where the optimal risky portfolio is a broad market index (market portfolio)



Review Questions

1. Treasury Bills are virtually risk free because they have

A) Long maturity and fixed interest rate

B) Long maturity and government guarantee

C) Short maturity and government guarantee

D) Short maturity and high interest rate



Review Questions

2. The capital allocation line (CAL) depicts all the risk-return combinations available to investors.

A: True

B: False



Review Questions

3. The slope of the Capital Allocation Line equals

- A) Excess risk for each unit of return
- B) Investors' risk aversion coefficient
- C) Certainty equivalent rate
- D) Excess return per unit of additional risk



Review Questions

4) Which portfolio would a risk-averse investor always choose ? A portfolio of T-bill with a rate of return of 5% and

A) Security A: $E(r) = 0.15$; Variance = 0.04

B) Security B: $E(r) = 0.10$; Variance = 0.0225

C) Security C: $E(r) = 0.12$; Variance = 0.01

D) Security D: $E(r) = 0.13$; Variance = 0.0625



Review Questions

5) You are considering risk free borrowing to increase the size of your portfolio. You will maintain the current asset allocation proportions. The leveraged portfolio will have a reward to variability ratio _____ than your current portfolio.

A) Higher

B) Lower

C) Can not be answered

D) Unchanged



Review Questions

	Large Stocks		1 month T bill	Risk Premium	Reward to Risk ratio
	Average	SD	Average		
1926 – 2002	12.04	20.81	3.82	8.22	0.4
1983 – 2002	13.98	16.26	5.60	8.38	0.5
1963 - 1982	9.67	16.90	6.61	3.06	0.18
1943 - 1962	16.56	18.14	1.48	15.07	0.83
1926 - 1942	7.23	29.58	1.20	6.03	0.20

- If your risk aversion coefficient is 4.
- You believe the 1926 – 2002 period is a representative of the future, then what fraction of your portfolio should be allocated to T bill and what fraction to equity ?
- What if you believe 1983 – 2002 is the representative period ?
- Comparing your answers, why do you think your capital allocation changes ?



Solution

Representative Period 1926 – 2002

$$A = 4, E(r_M) - r_f = 8.22\%, \sigma_M = 20.81\%$$

$$y^* = \frac{E(r_M) - r_f}{0.01A\sigma_M^2} = \frac{8.22}{0.01 \times 4 \times 20.81^2} = 0.4745$$

Equity = 47.45% and T-bills = 52.55%

Representative Period 1983 – 2002

$$A = 4, E(r_M) - r_f = 8.38\%, \sigma_M = 16.26\%$$

$$y^* = \frac{E(r_M) - r_f}{0.01A\sigma_M^2} = \frac{8.38}{0.01 \times 4 \times 16.26^2} = 0.7924$$

Equity = 79.24% and T-bills = 20.76%



Summary

- ❑ Capital Allocation – allocating funds between the optimal risky portfolio and the risk free asset
- ❑ T-Bill & T Bonds are proxies for risk-free asset
- ❑ Capital Allocation Line (CAL) represents all investment opportunities
- ❑ Capital allocation decisions are based on CAL and investor's utility function

